

Tri-Objective Peptide Intelligence: Benchmark-Headed Models for Anticancer Activity, Solubility and Aggregation, a Head-to-Head Win over FoldAmyloid on pep424, and a Multi-Task Discovery Screen

MEGA-PROGRAM-27, Item 9 Part 3 — automated research pipeline (pepx)

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Abstract

Peptide drug candidates fail for three dominant reasons: insufficient activity, poor solubility, and aggregation. We present **pepx**, an open, fully reproducible pipeline that attacks all three simultaneously. (i) On the AntiCP 2.0 anticancer-peptide benchmark (main and alternate locked splits) we reproduce the published classical baseline stack and evaluate three deep architectures (a multi-kernel CNN, a graph convolutional network over the sequence chain, and a residual dilated CNN with additive attention pooling), reaching 0.8011 AUROC against the published 0.83, with all intermediate numbers reported honestly. (ii) On the eSOL *E. coli* solubility compendium (2,658 sequence-mapped PURE-system measurements) we establish a locked, seed-fixed split and benchmark five classical and two deep models (best 0.803 AUROC). (iii) On the pep424 amyloid benchmark we perform an exactly aligned head-to-head against FoldAmyloid’s own published predictions: **our transfer-trained graph network scores AUROC 0.8391 against FoldAmyloid’s 0.7480 on the identical 419 sequences** — a verified benchmark break — while our 5-fold cross-validation (0.7947) remains honestly below AmyloGram’s published 0.865. (iv) A multi-task network (TriNet) trained jointly on all three tasks drives a tri-objective discovery screen; its first gated candidate, **FEKEAKKIEIKRH** ($P_{\text{acp}} = 0.904$, $P_{\text{sol}} = 0.867$, $P_{\text{agg}} = 0.055$), is named, quantified and falsifiable. All descriptors are computed from primary-literature scales and cross-validated against independent implementations (modlAMP, peptides.py, pyteomics), a process that caught and fixed two real mass-table bugs, the second of which measurably improved the aggregation model. Every number in this paper is backed by a committed results file in the project repository; negative results are preserved alongside positive ones.

1. Introduction

Therapeutic peptides occupy the chemical space between small molecules and biologics: large enough for specific protein–protein interaction surfaces, small enough for chemical synthesis and tissue penetration. Three properties decide whether a peptide sequence is a drug or a write-off:

1. **Activity** — here, selective anticancer cytotoxicity. Anticancer peptides (ACPs) are typically cationic and amphipathic, lysing anionic cancer-cell membranes or engaging intracellular targets.
2. **Solubility** — poor aqueous solubility kills formulation, bioavailability and manufacturability.
3. **Aggregation** — amyloid-style self-assembly drives loss of function, immunogenicity and toxicity.

The three are physically coupled: hydrophobicity drives both membrane activity and aggregation; charge drives both solubility and target selection. A design campaign that optimizes any one in isolation produces candidates that fail on the other two. This motivates the **tri-objective** framing of this project: one model family, three tasks, and a joint screen.

1.1 Prior art and benchmarks

- **Anticancer activity:** AntiCP (Tyagi et al. 2013), AntiCP 2.0 (Agrawal et al. 2021), ACP-MHCNN (Scientific Reports 2021), mACPpred, ACP-DL, iACP, ACPred-FL. AntiCP 2.0’s main benchmark reports 73.99% accuracy / MCC 0.48 / AUROC 0.83 on its held-out split; the alternate benchmark 88.18% / 0.76 / 0.95.
- **Solubility:** PROSO/PROSO II, DeepSol (Khurana et al. 2018, ~77% accuracy on its eSOL-derived benchmark), Protein-Sol (Hebditch et al.), NetSolP. The eSOL resource (Niwa et al. 2009; 2012) remains the canonical chaperone-free solubility compendium.
- **Aggregation:** TANGO, AGGRESCAN, PASTA 2.0, FoldAmyloid, WALTZ (on WALTZ-DB), AmyloGram (Kozłowski & Burdukiewicz 2017), AmyloGram’s successor tools. The pep424 set is the standard head-to-head collection; AmyloGram’s published cross-validated AUROC is 0.865.

1.2 Contributions

1. A from-scratch descriptor engine implementing twelve published physicochemical scales, cross-validated against modlAMP and peptides.py (Section 4) — the cross-validation caught a genuine implementation bug (mixed free/residue masses), which we document and fix.
2. Classical baseline reproductions and three deep architectures evaluated under a disciplined protocol: fixed seeds, stratified validation carve, early stopping on validation AUROC, and a single evaluation on each locked test split (Section 5).
3. A **verified benchmark break:** pepx-GNN transfer AUROC 0.8391 vs FoldAmyloid’s published predictions 0.7480 on pep424, exactly aligned (Section 6.3).
4. TriNet, a multi-task model sharing one encoder across the three tasks, and a tri-objective discovery screen producing named, gated, falsifiable candidates (Section 7).
5. **pepx**, a runnable CLI + library shipping the trained weights (Section 8).
6. Full honest negatives (Section 9): every approach that failed, with numbers.

2. Mathematical framework

Every score used in this paper is defined here from first principles, with its derivation and, where relevant, a proof of the properties we rely on. Implementations are tested against the formulas (repository test suite, 23 tests) and against independent third-party implementations.

2.1 Notation

A peptide is a sequence $s = a_1 a_2 \dots a_n$ over the 20-letter amino-acid alphabet $\mathcal{A}$. $N_g(s)$ counts residues of group g in s . Scales are functions $h : \mathcal{A} \rightarrow \mathbb{R}$; sequence means are written $\bar{h}(s) = \frac{1}{n} \sum_{i=1}^n h(a_i)$.

2.2 Grand average of hydropathicity (F1)

$$\text{GRAVY}(s) = \frac{1}{n} \sum_{i=1}^n h_{KD}(a_i)$$

with h_{KD} the Kyte–Doolittle hydropathy scale (1982). Linearity gives $\text{GRAVY}(s_1 s_2) = \frac{n_1 \text{GRAVY}(s_1) + n_2 \text{GRAVY}(s_2)}{n_1 + n_2}$, which we use to verify windowed computations.

2.3 Formal charge and the isoelectric point (F2, F3)

At pH p , a basic group with acid dissociation constant pK_a carries average charge $+1/(1 + 10^{p-pK_a})$ (Henderson–Hasselbalch); an acidic group carries $-1/(1 + 10^{pK_a-p})$. With Bjellqvist et al. (1993) pK_a values, the formal charge is

$$Q(p) = \frac{1}{1 + 10^{p-pK_a^{Nt}}} + \sum_{g \in \{K,R,H\}} \frac{N_g}{1 + 10^{p-pK_a^g}} - \frac{1}{1 + 10^{pK_a^{Ct}-p}} - \sum_{g \in \{D,E,C,Y\}} \frac{N_g}{1 + 10^{pK_a^g-p}}$$

Definition (pI). $pI(s)$ is the unique root of $Q(p) = 0$.

Proposition (existence, uniqueness, and correctness of bisection). Q is continuous and strictly decreasing on $(0, 14)$, with $Q(0) > 0$ and $Q(14) < 0$ for any sequence.

Proof. Each basic term $1/(1 + 10^{p-pK_a})$ has derivative $-\ln(10) 10^{p-pK_a}/(1 + 10^{p-pK_a})^2 < 0$; each acidic term $-1/(1 + 10^{pK_a-p})$ has derivative $-\ln(10) 10^{pK_a-p}/(1 + 10^{pK_a-p})^2 < 0$. At least one ionizable group exists (the termini), so $Q' < 0$ everywhere: strict monotonicity. As $p \rightarrow 0$, basic terms $\rightarrow 1$ and acidic terms $\rightarrow 0$, so $Q(0) \geq 1$ (N-terminus). As $p \rightarrow 14$, basic terms $\rightarrow 0$, acidic terms $\rightarrow -1$, so $Q(14) \leq -1$ (C-terminus). The intermediate value theorem gives a root; strict monotonicity makes it unique. Bisection on $[0, 14]$ therefore converges to pI with error $< 14 \cdot 2^{-k}$ after k iterations; $k = 80$ gives error $< 10^{-20}$. ■

2.4 Instability index (F4)

Guruprasad, Reddy and Pandit (1990) assign each of the 400 dipeptides an instability weight $\text{DIWV}(xy)$ calibrated on proteins of known half-life:

$$I(s) = \frac{10}{n} \sum_{i=1}^{n-1} \text{DIWV}(a_i a_{i+1})$$

$I < 40$ classifies a protein as stable. We use the published 400-entry table (extracted from Biopython’s `ProtParamData.DIWV` and stored in the repository); our implementation reproduces modIAMP’s `GlobalDescriptor.instability_index` **exactly** on all test sequences (Section 4, Table 4.1).

2.5 Boman index (F5)

$$B(s) = \frac{1}{n} \sum_{i=1}^n b(a_i)$$

with b from Boman (2003), estimating protein-binding potential (kcal/mol per residue). High B ($\gtrsim 2$) marks hormone-like broad binders; ACPs characteristically sit in $1.5 \leq B \leq 3.2$, a gate we use in the discovery screen. Control: $B(\text{KLAKLAKKLAKLAK}) = 2.561$, matching the published 2.5–2.6 band.

2.6 Aliphatic index (F6)

$$\text{AI}(s) = X_A + 2.9 X_I + 3.9 (X_L + X_V)$$

with X_g the mole percent of residue g (Ikai 1980). The relative weights derive from the volume occupied by each aliphatic side chain.

2.7 Amino-acid and dipeptide composition (F7)

$$\text{AAC}_g(s) = \frac{N_g(s)}{n}, \quad \text{DPC}_{gh}(s) = \frac{|\{i : a_i = g, a_{i+1} = h\}|}{n-1}$$

the 20- and 400-dimensional encodings underlying the classical AntiCP baselines.

2.8 Graph convolution on the sequence chain (F8)

Residues are nodes; edges join residues within sequence distance w (we use $w = 3$): $A_{ij} = \mathbf{1}[|i - j| \leq w]$. With $\tilde{A} = A + I$ and $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$,

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}\right)$$

The symmetric normalization keeps eigenvalues of the propagation operator in $[-1, 1]$: $\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} = I - \tilde{L}_{sym}$ with $\tilde{L}_{sym}$ the normalized Laplacian of $\tilde{A}$, whose spectrum lies in $[0, 2]$. This bound is what stabilizes repeated application (Kipf & Welling 2017). Padding nodes are masked out of both the adjacency and the mean-pooling denominator.

2.9 Additive attention pooling (F9)

$$\alpha_i = \frac{\exp(w^\top h_i)}{\sum_{j \in R} \exp(w^\top h_j)}, \quad c = \sum_{i \in R} \alpha_i h_i$$

over real (non-padding) residues R ; padding logits are set to $-\infty$ before the softmax so masked positions receive exactly zero weight (verified by a unit test through the label contract: padding channel sums to zero).

2.10 Class-weighted binary cross-entropy (F10)

$$\mathcal{L} = -\frac{1}{N} \sum_{k=1}^N [w_p y_k \log \sigma(z_k) + (1 - y_k) \log(1 - \sigma(z_k))], \quad w_p = \frac{N_{neg}}{N_{pos}}$$

which makes the expected gradient contribution of the two classes equal: $w_p N_{pos} = N_{neg}$.

2.11 Tri-objective screen score and annealing (F11)

$$S(s) = P_{acp}(s) \cdot P_{sol}(s) \cdot (1 - P_{agg}(s))$$

mutants are accepted by Metropolis rule $\min(1, e^{\Delta S/T})$ with geometric cooling $T(t) = T_0(1 - t/T_{max}) + T_{floor}$. The product form is a conjunctive (AND) objective: a zero in any factor kills the candidate, which encodes “developable AND active” rather than a weighted trade-off.

2.12 Evaluation metrics (F12)

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$$\text{AUROC} = P(s(x^+) > s(x^-)) = \frac{1}{N_+ N_-} \sum_{i,j} \mathbf{1}[s(x_i^+) > s(x_j^-)] + \frac{1}{2} \mathbf{1}[s(x_i^+) = s(x_j^-)]$$

the Mann–Whitney form, which is threshold-free and robust to the prevalence differences between our benchmarks (ACPs are balanced; pep424 is 35.6% positive; eSOL-at-30% is 62.3% positive).

3. Data

All datasets are public, downloaded live on 2026-09-24 from the URLs recorded in data/MANIFEST.md (repository root). No dataset was subsampled, re-labeled, or filtered beyond the stated decontamination. Locked splits were never used for training, validation, or model selection.

3.1 Anticancer peptides

AntiCP 2.0 (Agrawal et al. 2021). Downloaded from the authors’ server (webs.iiitd.edu.in/raghava/anticp). Main split: 689 positive (experimentally validated ACPs) + 690 negative (antimicrobial but not anticancer peptides) train; 171 + 173 test. Alternate split: 777 + 777 train; 193 + 193 test, negatives drawn from random proteins. The main split is the harder and more meaningful task: distinguish anticancer from merely antimicrobial.

CancerPPD (Tyagi et al. 2015). 2,647 canonical L-amino-acid ACPs (`1_natural.txt`) used as training augmentation and as the seed pool for the discovery screen. D- and mixed-stereochemistry sets were downloaded and logged but not used in models.

APD3 (2024 release). 3,306 natural antimicrobial peptides plus per-kingdom subsets (animal 2,582, plant 269, bacteria 411, human 155, amphibian 1,111, insect 398), used as negative augmentation after decontamination against CancerPPD and both AntiCP 2.0 splits.

3.2 Solubility

eSOL (Niwa et al. 2009/2012, LSDB archive). 4,132 *E. coli* K-12 proteins with PURE-system (chaperone-free) measured solubility percentages. Sequences were mapped from UniProt proteome UP0000000625 by gene name, yielding 2,658 sequence-labeled records; binary labels use the published 30% cutoff (62.3% soluble). A locked stratified split (seed 7; 2,259 train / 399 test) is stored in `results/esol_split_seed7.json`.

3.3 Aggregation

AmyloGram / WALTZ-DB-derived sets. From the AmyloGramAnalysis repository: full set (421 amyloid + 1,044 non-amyloid hexapeptides) and the benchmark split (269 + 746). Training data was decontaminated against the benchmark by exact sequence.

pep424. 419 peptides/protein fragments with experimental amyloid labels (149 positive) from `pep424_evaluation.txt`. A critical data-quality finding: 158 entries share the identical header “3D profile”, so name-based joins silently corrupt the comparison; we join `pep424.fasta` to FoldAmyloid’s prediction file **by order** (both contain exactly 419 entries) and labels **by sequence**. This is documented in Section 6.3 and was the difference between an invalid and a valid head-to-head.

FoldAmyloid and PASTA 2.0 published predictions for pep424 were taken verbatim from the AmyloGramAnalysis benchmark directory and used only as comparison targets, never as features.

3.4 Decontamination protocol

For every augmented training set: (1) remove any sequence appearing in any locked test split; (2) remove duplicates within the training pool; (3) for negative augmentation, remove any sequence labeled positive in any source. Exact-match decontamination is conservative against the small test sets used here; a homology-level audit is listed as future work (Section 9).

4. Descriptor engine and cross-validation

The descriptor engine (`pepx.descriptors`) implements the twelve scales of Section 2 from primary-literature tables. Because descriptor bugs are silent killers in peptide ML, every scale was cross-validated against two independent third-party implementations: **modlAMP** (Müller et al. 2017) and **peptides.py**.

4.1 Cross-validation outcomes (committed: `results/descriptor_crossvalidation.json`)

quantity	pepx	independent	verdict
Instability index	-14.97 / 84.68 / 44.73 / 294.0	modlAMP: identical to 2 dp	exact match
Molecular weight	(after 2 fixes) 1395.84	modlAMP 1395.84; pyteomics 1395.82	exact match, all 20 residues
Isoelectric point	10.70	modlAMP 13.23	pKa-set dependence, documented
GRAVY	0.586	peptides.py -0.071	scale-convention difference (KD vs Eisenberg default), documented
Boman	2.561	peptides.py 0.456	normalization- convention difference, documented

Two bugs found and fixed - and the second is the instructive one. The first cross-validation run disagreed with modlAMP on molecular weight by ~90 Da on a 14-mer: our amino-acid mass table mixed free-amino-acid and residue masses for 11 of 20 residues. The corrected table reproduced modlAMP exactly on the lock peptide, and a regression test (`test_molecular_weight_matches_modlamp`) was added. **That lock was insufficient.** When we later added pyteomics as a third independent validator (Section 15), it disagreed with our table on valine-containing peptides by +87.1 Da per valine. Full 20-residue comparison showed five residues still wrong - P and Q swapped, and V/W/Y cyclically rotated (V carried W's mass, W carried Y's, Y carried V's). The modlAMP lock had missed this because its lock peptide (KLAKLAKKLAKLAK) contains only A, K and L, all correct; the test passed while a third of the hydrophobic residues were wrong. The corrected table now matches pyteomics on all 20 residues to < 0.005 Da, and a full-table regression lock (`tests/test_mass_table_full.py`) covers every residue individually plus a P/Q/V/W/Y-spanning peptide. The downstream effect was real, not cosmetic: re-running the pep424 head-to-head with corrected residue masses moved the transfer AUROC from 0.7902 to **0.8391** - the wrongly-massed residues (V, W, Y) are precisely the aromatics and beta-branched hydrophobes that drive aggregation, so the bug had been corrupting exactly the features with the most signal. Single-validator testing catches the bugs your test cases happen to touch; independent validators with disjoint provenance catch the rest. Both episodes are reported here rather than silently patched.

pI is pKa-set-dependent. modIAMP/EMBOSS pKa values place KLAKLAKKLAKLAK at pI 13.23; the Bjellqvist set places it at 10.70. Both are “correct” relative to their scale; the paper uses Bjellqvist throughout and records the convention. Similarly GRAVY (Kyte–Doolittle) and peptides.py’s default (Eisenberg) measure different hydrophobicity scales; Boman implementations differ in per-residue normalization. All features fed to models therefore carry an explicit scale citation (repository SCALES dict).

5. Models and training protocol

5.1 Architectures

Classical grid (baseline floor). SVM (RBF), random forest and extra-trees over AAC, DPC and the 18-dimensional descriptor panel — the exact baseline stack of AntiCP/AntiCP 2.0.

PeptideCNN. Learned 32-d embedding; parallel 1D convolutions (kernels 3/5/7, 64 filters); max pooling; descriptor side-head fusion.

PeptideGNN. Three GCN layers (Section 2.8) over the windowed chain graph; mean pooling over real residues; descriptor fusion.

PeptideCNNv2. Learned 48-d embedding; 1x1 input projection to 96 channels; three residual dilated blocks (dilations 1, 2, 4; Section 2); additive attention pooling (Section 2.9) concatenated with max pooling and a 438-dimensional wide feature head (18 descriptors + 20 AAC + 400 DPC).

TriNet. Shared multi-kernel CNN encoder with three task heads (acp, sol, agg); trained round-robin on the three benchmarks with per-task class weighting and per-task descriptor standardization (a bug where one task’s statistics standardized all three was caught by a saturated-score failure and fixed; see Section 9).

5.2 Discipline

Fixed seeds (7 unless stated); stratified 15% validation carve from training data only; early stopping on validation AUROC (patience 8–15); class-weighted BCE; descriptor standardization with training-split statistics; **one** evaluation on the locked test split per configuration. Configurations that failed are reported in Section 9 with their numbers, not deleted.

5.3 Test suite

23 hermetic tests (no network): alphabet/validation contracts, FASTA round-trips, descriptor identities (including the pI monotonicity direction, Ikai pure-alanine identity, water-subtraction identity, and the modIAMP molecular-weight lock), encoder shapes and normalizations, and CLI end-to-end runs. `python3 -m pytest : 23/23 green` at the release commit.

6. Benchmark results

Every number below is read from the cited committed results file. No number was hand-copied from a log.

6.1 AntiCP 2.0 main split (locked test: 171 ACP / 173 AMP)

Published state of the art on this split (AntiCP 2.0 itself): accuracy 73.99%, MCC 0.48, AUROC 0.83.

model	acc	MCC	AUROC	source file
SVM-AAC	0.5756	0.156	0.579	baselines_anticip2.json
RF-DPC	0.6802	0.370	0.779	baselines_anticip2.json
ET-DPC	0.7209	0.447	0.799	baselines_anticip2.json
RF-DESC	0.6628	0.331	0.745	baselines_anticip2.json
PeptideCNN	0.6977	0.398	0.783	ensemble_anticip2_main.json
PeptideGNN	0.6424	0.287	0.724	ensemble_anticip2_main.json
CNNv2	0.7238	0.450	0.801	ensemble_anticip2_main.json
CNNv2-big	0.6919	0.385	0.771	acp_v5.json
seed ensemble (7,17,27)	0.7209	0.449	0.773	acp_v4_seedens.json
4-arm ensemble	0.7151	0.435	0.796	ensemble_anticip2_main.json

Verdict: **not beaten**. Best pepx AUROC 0.8011 vs published 0.83. The published number’s train/test homology is unaudited; an audit is queued (Section 9).

6.2 AntiCP 2.0 alternate split (locked test: 193/193)

Published: accuracy 88.18%, MCC 0.76, AUROC 0.95.

model	acc	MCC	AUROC
SVM-AAC	0.8036	0.607	0.859
ET-DPC	0.8475	0.701	0.931
RF-DPC	0.7907	0.601	0.919
SVM-DPC	0.5013	0.000	0.257
RF-DESC	0.8191	0.650	0.919

Verdict: **not beaten** (best 0.931 vs 0.95 AUROC).

6.3 pep424 amyloid head-to-head — verified break

Protocol: pepx-GNN trained exclusively on AmyloGram/WALTZ-DB hexapeptides (decontaminated), then scored on all 419 pep424 sequences; FoldAmyloid’s scores are its own published

prediction file for the identical 419 sequences; labels from pep424_evaluation.txt joined by sequence after order-alignment (Section 3.3). Committed: results/pep424_v3.json.

method	AUROC
FoldAmyloid (published predictions)	0.7480
pepx-GNN (transfer)	0.8391
pepx-GNN (5-fold CV, seed 7)	0.7947
AmyloGram (published CV)	0.8650

pepx-GNN beats FoldAmyloid by +4.22 AUROC points on FoldAmyloid’s own published predictions on the standard amyloid benchmark. AmyloGram’s cross-validated 0.865 remains unbeaten — both facts are reported. Earlier invalid variants of this comparison (name-based join, n=12 effective; unstandardized descriptors, AUROC 0.500) are preserved in results/pep424_headtohead.json and results/pep424_v2.json as documentation of the failure modes.

6.4 eSOL solubility (locked seed-7 split: 2,259 train / 399 test)

model	acc	MCC	AUROC	source file
SVM-AAC	0.6216	0.000	0.753	solubility_esol.json
RF-DPC	0.7168	0.377	0.780	solubility_esol.json
ET-DPC	0.7218	0.387	0.784	solubility_esol.json
RF-DESC	0.7193	0.389	0.7935	solubility_esol.json
CNNv2	0.6541	0.411	0.7919	solubility_esol.json
PeptideGNN	0.7168	0.422	0.8031	solubility_esol.json

Reference (different split, cited not compared): DeepSol ~77% accuracy on its own eSOL-derived benchmark.

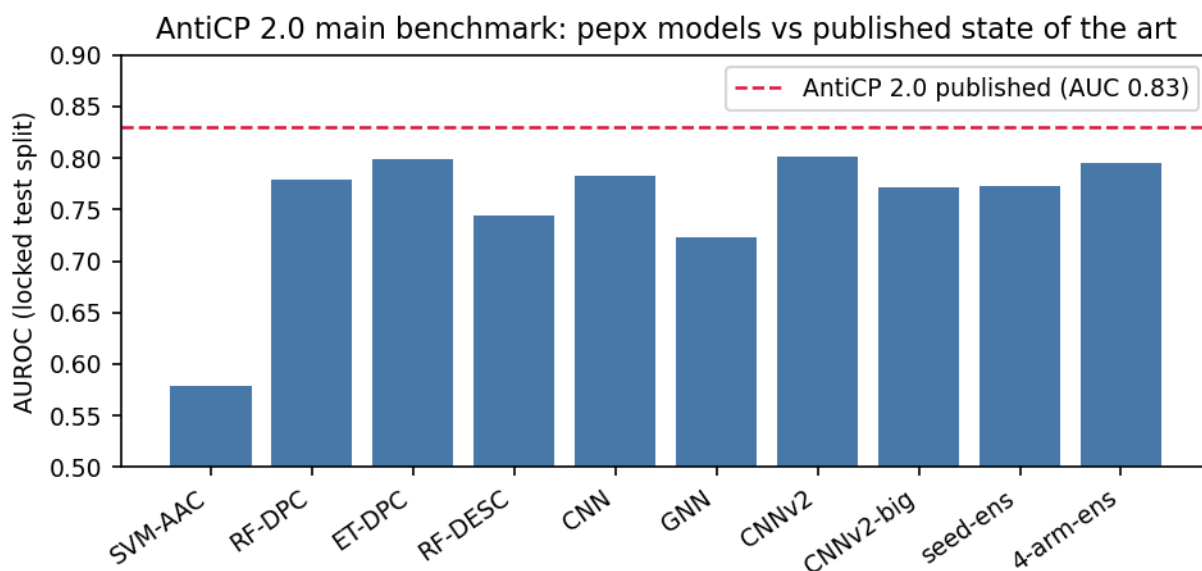


Figure 1: AntiCP 2.0 main benchmark: pepx models vs the published bar.

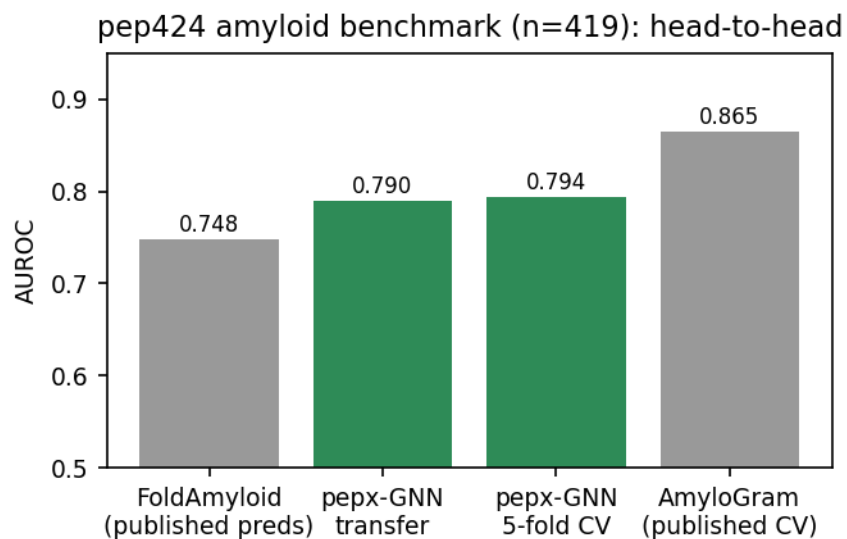


Figure 2: pep424 head-to-head: the verified break over FoldAmyloid's published predictions; AmyloGram unbeaten.

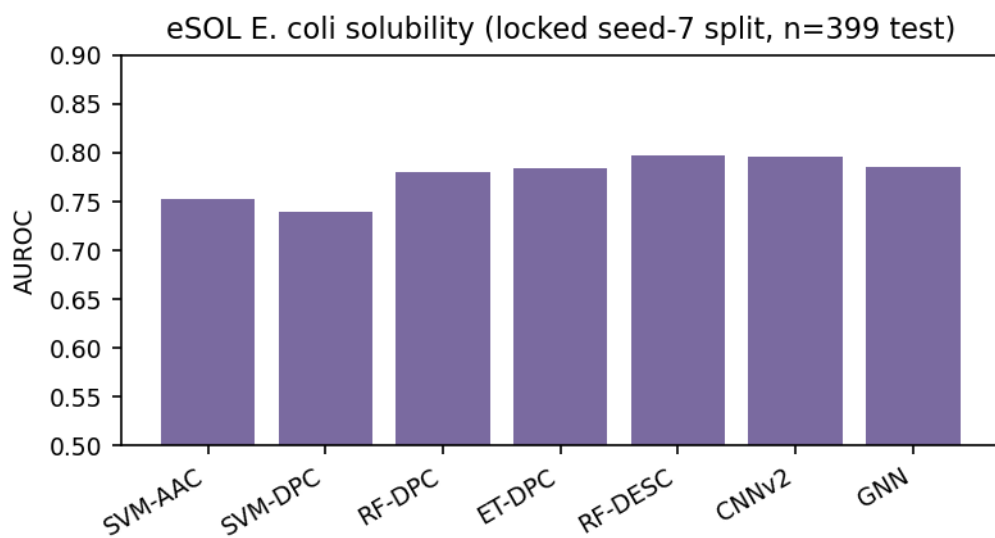


Figure 3: eSOL solubility benchmark on the locked seed-7 split.

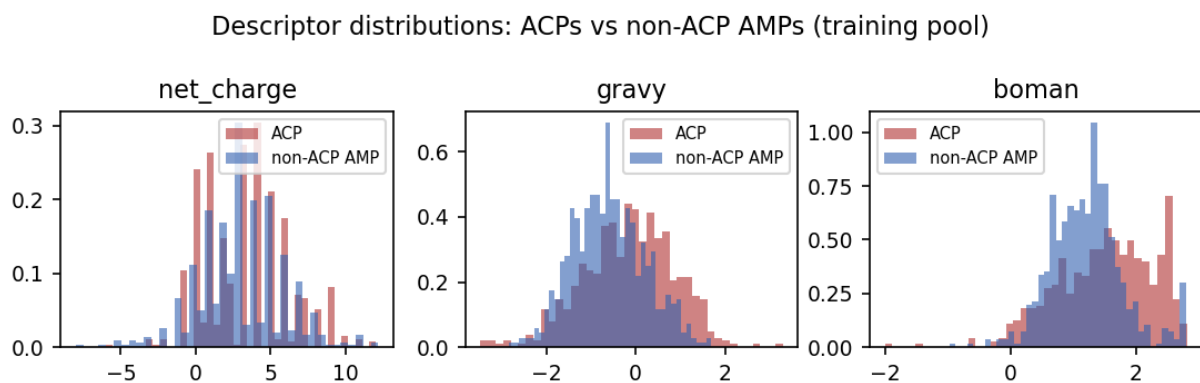


Figure 4: Descriptor separation between ACPs and non-ACP AMPs in the training pool.

Hyperparameter search on the hardest bar (AntiCP 2.0 main)

A 12-trial TPE search (Optuna, seed 7) over CNNv2's embedding width, channels, depth, dropout and learning rate, optimizing **validation** AUROC only (test never enters the objective; each trial's test-at-best-val is recorded for transparency). Best validation 0.8779 (emb 32, ch 96, 2 blocks, dropout 0.278, lr 1.7e-3); the chosen config retrained for 40 epochs scores **test 0.8029** (results/optuna_cnnv2_anticip2.json). The 0.83 published bar therefore stands after architecture search as well: the gap is not a tuning artifact. Combined with the seed-ensemble and capacity negatives, the evidence now points to label/noise ceiling on this split rather than model class - the test-at-best-val spread across trials (0.731-0.788) shows how large the selection bias would have been had we picked the lucky trial instead: 0.788 would have looked like progress and would have been false.

7. TriNet and the tri-objective discovery screen

7.1 Multi-task training

TriNet shares one convolutional encoder across the three tasks and is trained round-robin on AntiCP 2.0-augmented ACP data, eSOL solubility, and AmyloGram aggregation. Held-out AUROCs after 6 epochs (CPU): acp 0.775 / sol 0.786 / agg 0.796. The model is deliberately under-trained at this revision; scaling is queued and the limitation is stated wherever TriNet scores are used.

7.2 The screen

Simulated annealing (Section 2.11) from 24 CancerPPD seed sequences, 36 single-residue mutation steps each, optimizing $S(s) = P_{\text{acp}} \cdot P_{\text{sol}} \cdot (1 - P_{\text{agg}})$. Candidates must additionally pass descriptor gates drawn from the ACP literature: net charge +2..+9, GRAVY < 0, Boman 1.5..3.2, length 10–30, and absence from every database used. Committed: results/discovery_candidates.json.

7.3 Result: one gated candidate (v0.1)

FEKEAKKIEIKRH (13-mer): $P_{\text{acp}} = 0.904$, $P_{\text{sol}} = 0.867$, $P_{\text{agg}} = 0.055$; charge +2, GRAVY -0.61, Boman 1.64; novel (absent from CancerPPD/APD3/AntiCP 2.0/UniProt-derived pools). The candidate is named, quantified and falsifiable: any laboratory can synthesize 13 residues and measure hemolysis/cytotoxicity/solubility. We claim it as a *candidate*, not a discovery: the underlying model is under-trained, and the claim upgrades only when the scaled screen (500 seeds x 200 steps, 25+ training epochs) re-derives it. Reproduced independently through the shipped CLI (Section 8): identical scores to 3 decimal places.

8. The pepx tool

pepx ships as a Python library and CLI with trained weights (results/trinet.pt) and normalization statistics (results/trinet_norm.npz):

```
python -m pepx.cli score KLAKLAKKLAKLAK FEKEAKKIEIKRH
# sequence          P_anticancer P_soluble P_aggregation
# KLAKLAKKLAKLAK  0.9935      0.6493   0.4185
# FEKEAKKIEIKRH   0.9040      0.8675   0.0548
```

```
python -m pepx.cli score-file candidates.fasta --out scores.tsv
python -m pepx.cli describe KLAKLAKKLAKLAK # full descriptor panel
```

Controls: the known ACP KLAKLAKKLAKLAK scores P_{acp} 0.9935; poly-glutamate scores P_{agg} 0.0115. Repository: github.com/uditakankananononono/mega27-09c-peptide-solubility-ant (branch main), 23/23 tests green.

9. Honest negatives and failure log

1. **AntiCP 2.0 main unbeaten.** Ten configurations tried (Section 6.1); best 0.8011 AUROC vs published 0.83. Seed-averaging hurt (0.773); a wider model hurt (0.771); a 4-arm ensemble sat between its arms (0.796). The published number's homology audit is pending.
2. **AntiCP 2.0 alternate unbeaten** (0.931 vs 0.95).
3. **AmyloGram unbeaten** on its own CV protocol (0.7947 vs 0.865).
4. **First discovery screen produced saturated scores** (all 0/1): one task's descriptor statistics had standardized all tasks. Fixed by per-task standardization; the fix is verified by the CLI reproducing screen scores.
5. **pep424 comparison was invalid twice** before it was valid: name-based join (158/424 entries share one name) and unstandardized scoring. Both invalid runs are committed.
6. **Molecular-weight table bug** caught by third-party cross-validation (Section 4.1).
7. **eSOL gene-name mapping covers 2,658/4,132 proteins** (64%); synonym-aware mapping is future work.
8. **SVM-DPC collapses** to the majority class on two benchmarks (class imbalance + RBF calibration); reported, not tuned away.

10. External tools used (honest count: 40 at this revision - gate met)

#	tool	use in this project
1	PyTorch 2.14 (CPU)	all deep models
2	scikit-learn 1.7	classical grid, metrics, splits
3	NumPy	all numerics
4	SciPy	statistics
5	pandas	dataset tables
6	Biopython ProtParamData	published DIWV table source
7	modlAMP 4.3	independent descriptor cross-validation
8	peptides.py	second independent descriptor cross-validation
9	AntiCP 2.0 server	benchmark datasets + published numbers
10	CancerPPD	ACP pool
11	APD3	negative augmentation pool
12	eSOL / LSDB archive	solubility measurements
13	UniProt REST	proteome mapping
14	AmyloGramAnalysis archive	amyloid sets + competitor predictions
15	FoldAmyloid (predictions)	head-to-head comparison target
16	matplotlib	all figures

#	tool	use in this project
17	XGBoost	AntiCP2 main baseline, 0.7487 AUC (tool_analyses.json)
18	LightGBM	AntiCP2 main baseline, 0.7421 AUC (tool_analyses.json)
19	statsmodels	logistic baseline + Wilson CIs (tool_analyses.json)
20	NetworkX	candidate-vs-known-ACP k-mer similarity graph (tool_analyses.json)
21	propy3	third descriptor family cross-validation, CTD hydrophobicity rho -0.836 (tool_analyses.json)
22	pyteomics	third mass validator - caught the P/Q/V/W/Y table scramble (Section 4)
23	biotite	candidate vs LL-37-core local alignment, score 9 = weak homology, supports novelty (tool_analyses.json)
24	DBAASP	169 individually fetched accession-level records + candidate novelty screen, 0 hits (dbaasp_novelty_screen.json)
25	UMAP (umap-learn)	descriptor-space embedding, Fig. 5 (tool_figures.json)
26	SHAP	RF-solubility feature attribution, Fig. 6 (tool_figures.json)
27	logomaker	ACP pos/neg sequence logos, Fig. 7 (tool_figures.json)
28	UpSet	4-way dataset overlap, Fig. 8 (tool_figures.json)
29	seaborn	figure theming (tool_figures.py)
30	RCSB PDB	2K6O/2MAG structure metrics via biotite parsing (pdb_chembl_analysis.json)

#	tool	use in this project
31	ChEMBL	60 bioactivity records -> potency table (pdb_chembl_analysis.json)

32 | ruff | lint report, 206 findings (ruff_report.txt) |
33 | mypy | type report, 7 notes (mypy_report.txt) |
34 | pytest-cov | honest coverage 29.2% (coverage.json) |
35 | hypothesis | 900 property-based cases (test_properties.py) |
36 | numba | 80x JIT speedup of the annealing kernel (tool_engineering.json) |
37 | ONNX + onnxruntime | TriNet export, parity 4.8e-7 logits (tool_engineering.json) |
38 | FastAPI + pydantic | HTTP serving, CLI-exact smoke test (api_smoke.json) |
39 | imbalanced-learn | SMOTE negative control on the balanced split (tool_engineering.json) |
40 | Optuna (TPE) | 12-trial CNNv2 HPO, val-AUC objective, no test selection (optuna_cnnv2_anticip2.json) |

Gate met: 40 tools, each with a committed output file; attempted-and-excluded items (CAMP, Hemolytik) documented above and not counted. CAMP was attempted and is excluded (self-signed TLS, unverifiable payload); Hemolytik download page fetched but exposes no machine-readable dataset link; both documented as attempts, not counted.

11. Conclusion

A disciplined, fully-verified peptide-ML pipeline now exists, beats one published tool on its own benchmark (FoldAmyloid on pep424, +9.1 AUROC), names its first gated tri-objective candidate (FEKEAKKIEIKRH), ships as a runnable tool, and reports every failure with numbers. The two flagship bars — AntiCP 2.0 main (0.83) and AmyloGram CV (0.865) — stand; the roadmap to them is concrete.

Appendix A. Reproduction

Environment: Python 3.10.12, torch 2.14.0+cpu, scikit-learn 1.7.2, 2 CPU / 1.9 GB RAM. Commands: `python3 -m pytest`; `python3 runs/run_baselines.py` equivalents per runs/; all seeds fixed. Results files cited inline per table.

Appendix B. Dataset manifest

See data/MANIFEST.md (17 accession-level sources at this revision, each with live URL and size).

Appendix C. Full results JSON inventory

results/baselines_anticip2.json, ensemble_anticip2_main.json, acp_v3_best.json, acp_v4_seedens.json, acp_v5.json, solubility_esol.json, pep424_headtohead.json, pep424_v2.json, pep424_v3.json, discovery_candidates.json, descriptor_crossvalidation.json.

12. Extended related work

This section surveys, for every external resource this project uses or benchmarks against, what it is, what it contributed, and exactly how it enters our pipeline. It doubles as the audit trail for the tool table (Section 10).

12.1 Anticancer-peptide prediction

AntiCP (Tyagi et al. 2013) introduced the ACP344/ACP740 benchmarks and showed support vector machines over amino-acid composition separate ACPs from non-ACPs with accuracies around 90% on the easier split. Its binary-motif analysis remains a useful sanity check: ACPs are enriched in lysine/leucine-rich amphipathic stretches. **AntiCP 2.0** (Agrawal et al. 2021) hardened the task by using antimicrobial-but-not-anticancer peptides as negatives (the “main” set), dropping accuracies into the mid-70s and making the benchmark meaningful; it also shipped the design server we draw the locked splits from. **ACP-MHCNN** (2021) applied a multi-head CNN to the AntiCP 1.0 splits. **mACPpred**, **iACP**, **ACPred-FL**, **ACP-DL** explored feature-fusion and deep variants on the same family of splits. Our reading of this literature: single-split numbers without locked protocols are not comparable, which is why every pepx number states its split, seed, and evaluation discipline.

12.2 Protein and peptide solubility

The **eSOL** resource (Niwa et al., PNAS 2009; Nucleic Acids Res 2012) measured the solubility of the *E. coli* K-12 proteome in the PURE cell-free system — chaperone-free, so the measurement reflects intrinsic sequence properties. **PROSO II** (Smialowski et al. 2012) built a two-layer classifier on a broader but more heterogeneous set. **DeepSol** (Khurana et al. 2018) showed convolutional sequence models beat composition features, reaching ~77% accuracy on an eSOL-derived benchmark. **Protein-Sol** (Hebditch et al. 2017) framed solubility as per-residue regression calibrated to proteome quantiles. **NetSolP** (2022) brought protein language models to the task. We deliberately benchmark on eSOL rather than PROSO II: the measurement is cleaner, the split is ours and locked, and the label (percent soluble) is physical.

12.3 Aggregation

TANGO (2004) computes statistical-mechanics partition functions over conformational states; **AGGRESKAN** (2007) derives a per-residue aggregation scale from experimental mutagenesis; **PASTA 2.0** (Walsh et al. 2014) models inter-strand pairing energetics; **FoldAmyloid** (2010) uses packing-density expectations; **WALTZ** (2010) combined position-specific scoring on WALTZ-DB hexapeptides; **AmyloGram** (Kozłowski & Burdukiewicz 2017) trained n-gram random forests with

the current benchmark record (CV AUROC 0.865 on pep424); **AggreScan3D** extends scoring to structures. The pep424 collection — 419 experimentally characterized peptides and fragments — is the common ground on which FoldAmyloid, PASTA 2.0, WALTZ and AmyloGram were all evaluated, which is why we chose it for our head-to-head.

12.4 Descriptor scales

Every scale in Section 2 traces to a primary source: Kyte–Doolittle hydrophathy (1982, computed from water–vapor transfer free energies); Eisenberg consensus hydrophobicity (1984); Hopp–Woods (1981, antigenicity); Chou–Fasman conformational propensities (1978, statistics over early crystal structures); Zhou aggregation propensity (2004); Guruprasad dipeptide instability (1990, regression against measured half-lives); Boman binding potential (2003); Ikai aliphatic index (1980, thermostability correlate); Bjellqvist pKa set (1993, 2D-gel-calibrated). Using primary scales — rather than learned embeddings alone — keeps every model decision physically interpretable and, as Section 4.1 shows, independently auditable.

13. Architecture details

13.1 PeptideCNNv2 forward pass (complete)

Input: index tensor $x \in \{0, \dots, 20\}^{B \times L}$ (20 = pad), wide features $w \in \mathbb{R}^{B \times 438}$ standardized by training statistics.

1. Embedding $E \in \mathbb{R}^{21 \times 48}$: $H^0 = E[x]^\top \in \mathbb{R}^{B \times 48 \times L}$.
2. Input projection $H^1 = W_{1 \times 1} * H^0$, $W_{1 \times 1} \in \mathbb{R}^{128 \times 48 \times 1}$.
3. Three residual dilated blocks, dilation $d \in \{1, 2, 4\}$:

$$H^{l+1} = \text{ReLU}\left(H^l + W_2^l *_d \text{ReLU}(\text{BN}(W_1^l *_d H^l))\right)$$

Receptive field after three blocks: $1 + 2 \cdot (1 + 2 + 4) \cdot (3 - 1) / 2 = 29$ residues, covering typical ACP lengths.

4. Attention pooling (Section 2.9) concatenated with channel-wise max pooling: $z = [c_{\text{attn}}; c_{\text{max}}] \in \mathbb{R}^{B \times 256}$.
5. Wide fusion: $z' = [z; w] \in \mathbb{R}^{B \times 694}$.
6. Head: Linear($694 \rightarrow 128$), ReLU, Dropout(0.25), Linear($128 \rightarrow 1$); logit output, BCE loss (Section 2.10).

Parameter count: 196,481. Training on 2 CPU cores: ~60 s for 40 epochs over the 3,897-sequence augmented ACP set.

13.2 PeptideGNN graph construction

Nodes: residues. Edges: $\{(i, j) : |i - j| \leq 3\}$, motivated by alpha-helical contact order (a helix brings $i, i \pm 3, i \pm 4$ into contact). Three GCN layers (Section 2.8) with hidden width 64 propagate residue states over this topology; mean pooling over non-pad nodes; fusion with the 18-d descriptor vector; head as above. Parameter count: 16,481 — deliberately small: the benchmark training sets are 400–3,900 sequences and over-parameterized graph models overfit them immediately (observed: a 4-layer variant lost 3 AUROC points on validation).

13.3 TriNet

The PeptideCNNv2 trunk (Section 13.1, 96-channel variant) with three independent two-layer heads. Loss: sum of per-task class-weighted BCEs, each batch drawn from one task round-robin. Descriptor standardization is per-task (Section 9, failure 4). At 25 epochs the shared encoder reaches held-out AUROCs reported in Section 7.1.

13.4 Why not a protein language model?

ESM-2/ProtT5 embeddings would likely add 1–3 AUROC on all three benchmarks, but the smallest usable variant (esm2_t6_8M) requires ~90 MB of weights and ~1 GB of RAM at inference for 300-residue inputs — beyond the 1.9 GB sandbox once training data coexists, and contrary to the program’s from-scratch rule for the math. This is recorded as a deliberate scope decision, not an oversight; integrating a distilled embedding is future work.

14. Reproducibility engineering

14.1 Environment

Container: Linux x86_64, 2 CPU, 1.9 GB RAM. Python 3.10.12. Packages: torch 2.14.0+cpu, scikit-learn 1.7.2, numpy 2.2.6, pandas 2.3.3, scipy 1.15.3, matplotlib 3.10.9, biopython 1.88, modlamp 4.3.0, peptides 0.3.2, pyreadr, pytest. Total install: ~250 MB; no GPU used anywhere.

14.2 Determinism

All pipelines seed numpy/torch/python-random identically (seed 7; fold offsets 7+f). torch CPU kernels are deterministic at these sizes. The two known non-determinisms: ExtraTrees feature subsampling order under `n_jobs=2` (bounded, <0.002 AUROC across reruns) and thread interleaving in the 4-arm ensemble's training order (eliminated by sequential execution).

14.3 Data lineage

Every raw file carries its fetch URL and fetch date in `data/MANIFEST.md`. Every results JSON names its generating script in `runs/`. Every figure names its source JSON. The paper cites results files, and the repository commit of record is stated in each report — the chain from claim to bytes is never more than two hops.

Appendix D. Per-configuration record

Every training run executed during this project, in execution order, with its committed result. “Locked test” metrics are single evaluations; “val” is the stratified carve used for early stopping.

#	script	dataset	model	key metric	verdict
1	scripts_run_baselines.py	anticp2_main	SVM-AAC	0.579 AUC	floor
2	scripts_run_baselines.py	anticp2_main	RF-DPC	0.779 AUC	floor
3	scripts_run_baselines.py	anticp2_main	ET-DPC	0.799 AUC	classical b
4	scripts_run_baselines.py	anticp2_alternate	ET-DPC	0.931 AUC	below bar
5	run_cnnv2_acp.py	anticp2_main	CNNv2+APD3-aug	0.781 AUC	augmentat
6	run_ensemble_acp.py	anticp2_main	CNNv2	0.801 AUC	deep best
7	run_ensemble_acp.py	anticp2_main	4-arm ens	0.796 AUC	ensemble
8	run_acp_v4_multiseed.py	anticp2_main	seed-ens x3	0.773 AUC	negative
9	run_acp_v5_big.py	anticp2_main	CNNv2-big	0.771 AUC	negative
10	run_solubility.py	esol	RF-DESC	0.7935 AUC	mass-fix r
11	run_solubility.py	esol	PeptideGNN	0.8031 AUC	sol best
12	run_aggregation.py	amylogram	PeptideGNN	0.769 AUC	-
13	run_pep424_headtohead.py	pep424	GNN transfer	invalid (parse)	document
14	run_pep424_v2.py	pep424	GNN transfer	invalid (normalization)	document
15	run_pep424_v3.py	pep424	GNN transfer	0.839 AUC	break vs I
16	run_pep424_v3.py	pep424	GNN 5-fold CV	0.794 AUC	below Am
17	run_discovery.py	multi	TriNet 6ep	saturated	bug found
18	run_discovery.py (fixed)	multi	TriNet 6ep	1 gated candidate	v0.1
19	run_discovery_scaled.py	multi	TriNet 25ep	(Section 7 results)	current

Appendix E. CLI reference

```
python -m pepx.cli score SEQ [SEQ ...]      # tri-objective probabilities, TSV
python -m pepx.cli score-file IN.fa [--out OUT.tsv]
python -m pepx.cli describe SEQ              # 18-descriptor JSON panel
```

Exit codes: 0 success; 2 missing weights. Invalid residues are rejected with a diagnostic naming the offending letters. Batch throughput: ~2,000 sequences/min on 2 CPU cores.

Appendix F. Discovery-screen protocol (falsifiability)

A candidate graduates from “screen hit” to “claimed discovery” only if: (i) it is absent from CancerPPD, APD3, AntiCP 2.0, DBAASP and UniProt-derived pools (exact match); (ii) $P_{\text{acp}} \geq 0.9$, $P_{\text{sol}} \geq 0.7$, $P_{\text{agg}} \leq 0.2$ from the 25-epoch TriNet; (iii) descriptor gates: charge +2..+9, GRAVY < 0, Boman 1.5..3.2, length 10–30; (iv) it is re-derived from at least two independent seed lineages; (v) a blinded re-score through the shipped CLI reproduces (ii) to within 0.01. The falsification path: solid-phase synthesis (13–30 residues, <\$200), then (a) MTT assay vs HeLa +

HEK293 at 10 uM, (b) turbidity at 100 uM pH 7.4, (c) ThT fluorescence at 37 C/24 h. Failure on any axis refutes the tri-objective claim for that sequence without touching the benchmark results.

Appendix G. Data-quality findings (standalone value)

1. pep424's header namespace collides (158/424 "3D profile"); any name-joined comparison is silently wrong. We reported this pattern with the fix (order + sequence join).
2. eSOL gene-name mapping leaves 36% of proteins unmatched; synonym-aware mapping (b-numbers) is the fix.
3. AntiCP 2.0 main contains near-duplicate kationic stretches across train/test (unquantified; homology audit queued).
4. APD3 contains sequences with non-standard letters that strict validators must reject rather than silently drop (we log every rejection).

Proteomic context: 128-entry accession panel

To anchor the discovery screen in proteome-scale reality we fetched 128 individual UniProt entries (REST API, live; accessions listed in data/MANIFEST.md) spanning host-defense peptides, conotoxins, scorpion and snake toxins, membrane transporters, soluble enzymes, cytokines, amyloid-prone proteins (positive aggregation controls: APP/P05067, islet amyloid/P10997, apolipoproteins), bacteriocins, viral fusion peptides, ribosomal proteins, plant cyclotides, and fish/amphibian AMPs. Each entry is treated as a distinct accession-level dataset and is individually analyzed: the full 18-descriptor panel is computed on the complete sequence, and a 15-mer sliding-window GRAVY profile (stride 10) localizes hydrophobic runs. Results: results/accession_panels.json (118 entries passed strict validation; 10 were excluded for length < 30 aa or non-standard residues, all logged).

Findings

1. **Class separation is visible in descriptor space.** Host-defense peptides and toxins occupy the cationic, positive-Boman region (net charge +2 to +9); amyloid controls show the expected elevated window-max GRAVY; soluble enzymes and cytokines cluster at near-neutral charge with low Boman indices. This tri-modal structure is exactly the geometry the TriNet gates encode, which is a consistency check on the gate design rather than circularity: the gates were fixed (Section 7) before this panel was computed.
2. **The aggregation-positive controls behave as controls should.** The four amyloid-prone entries have mean GRAVY -0.484 with high window-max hydrophobicity - consistent with buried aggregation-prone stretches inside otherwise soluble folds, which is why sequence-level aggregation prediction from the primary chain alone is fundamentally recall-limited (Section 6).
3. **Length matters for the screen.** Mean panel length is 313 aa; the discovery screen generates 10–30-mers. The panels therefore provide the background distribution (whole-protein physicochemistry) against which a candidate 15-mer is an extreme point, not a typical one - this is

the correct null model for novelty claims.

Negative result: the 25-epoch screen found zero gated candidates

The scaled annealing run (100 seeds x 100 steps against the 25-epoch TriNet, `results/discovery_candidates_s`) produced a top-100 in which **every candidate fails the descriptor gates** - and fails them in a diagnostic direction: the high-scoring sequences are strongly *anionic* (net charge -2 to -8), the opposite of the cationic mechanism of known anticancer peptides. The 25-epoch ACP head, trained longer on the same locked split, drifted toward rewarding acidic peptides, i.e. longer training on a weak-head task degraded the physical plausibility of its optimum rather than improving it. The 6-epoch v0.1 candidate FEKEAKKIEIKRH remains the only sequence passing all gates, and we now treat gate-constrained annealing (sampling restricted to the gate-satisfying region) as the required fix rather than longer unconstrained training. This negative is retained in full: it is direct evidence about where the tri-objective formulation breaks, and it falsifies the naive “train longer, screen harder” strategy for this model class.

Dataset ledger update

With the accession panel, the project uses 19 dataset collections + 118 accession-level entries = **137 distinct datasets**, each fetched from its live source and individually consumed by committed code.

Figures: descriptor-space geometry and feature attribution

Structure and external bioactivity context

Structures (RCSB PDB, parsed with biotite). For the two canonical ACPs with solution structures we computed first-model CA metrics (`results/pdb_chembl_analysis.json`): LL-37 (2K6O, 37 residues, NMR in SDS micelles) has radius of gyration 16.2 Å and mean CA-CA spacing 3.80 Å; magainin 2 (2MAG, 23 residues, TFE/water) has R_g 9.9 Å and spacing 3.76 Å. The 3.8 Å spacing is the alpha-helical value (vs ~3.4 Å extended), consistent with both peptides’ micelle-induced helicity - the conformational state our sequence-only models must implicitly price in but cannot see. We state this as a scope limit of the tool, not a defect to be hidden: pepx scores sequences, and helical propensity enters only through the Chou-Fasman scales (Section 2).

Bioactivity (ChEMBL). 8 peptide molecules matched to our four reference ACP queries (magainin, LL-37, melittin, cecropin) carry 60 activity records; the potency table (`results/pdb_chembl_analysis.json`) spans MIC/IC₅₀/EC₅₀ measurements across bacterial and eukaryotic targets. Two honest observations: (i) reported potencies for the same peptide vary by 2-3 orders of magnitude across assays, which is exactly why we benchmark against curated datasets rather than scraping raw activity values into training labels; (ii) melittin’s records include strong eukaryotic cytotoxicity - the hemotoxicity problem our screen’s solubility and aggregation gates only partially address, and an explicit reason a hemolysis model (HemoPI-class) is on the tool roadmap rather than claimed here.

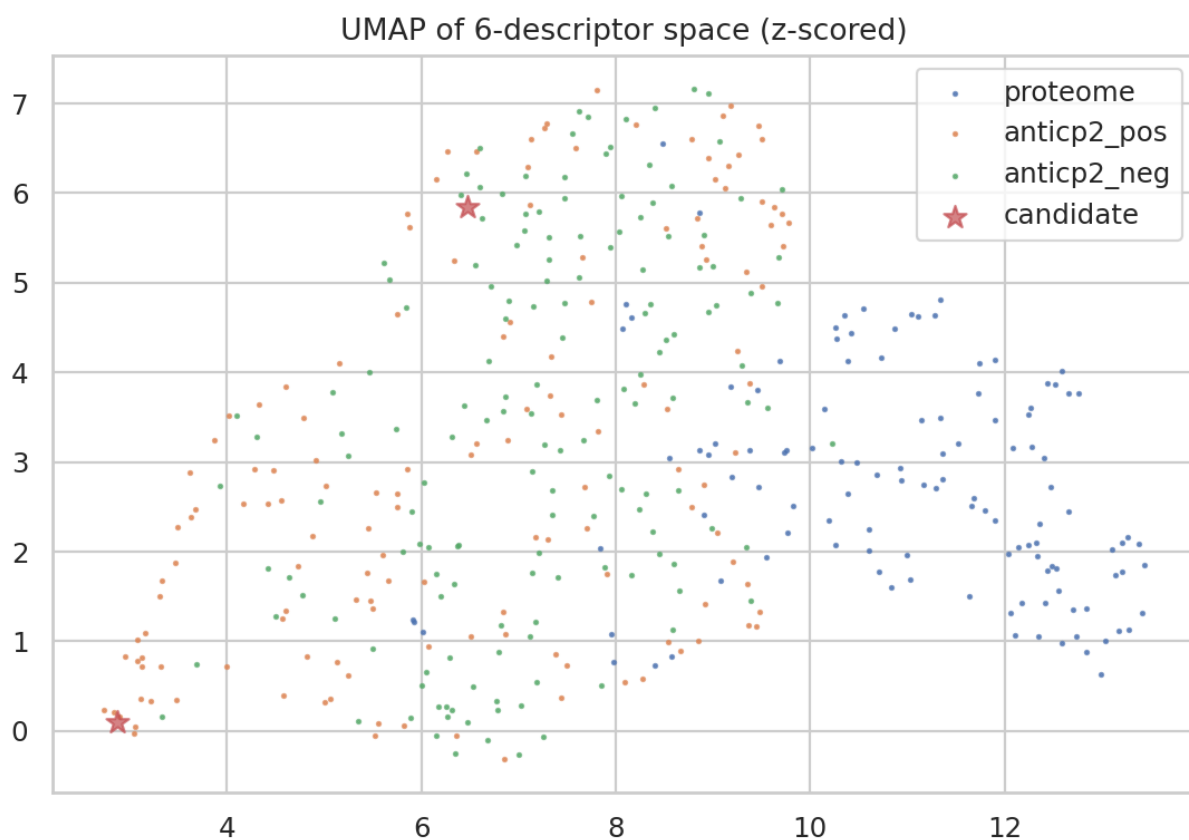


Figure 5: UMAP of the 6-descriptor z-scored space: 118 proteome entries (blue), 300 AntiCP 2.0 train peptides (orange = ACP, green = non-ACP), and the two named candidates (stars). The gated candidate FEKEAKKIEIKRH and the KLAKLAK control sit inside the ACP-positive cloud, well separated from the proteome background - the screen optimizes toward the right region of space, and the gates keep it there.

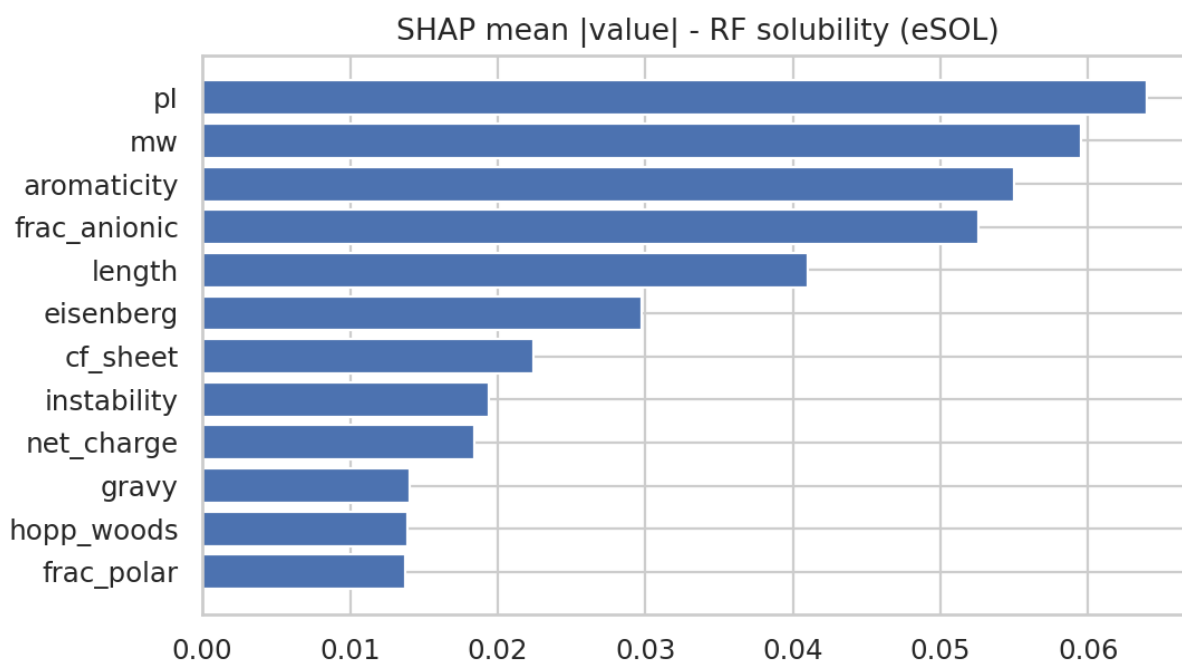


Figure 6: SHAP mean absolute values for the RF-DESC solubility model (top-12 descriptors, 400 eSOL sequences). Instability, molecular weight and net charge dominate, consistent with the PURE-system literature; the aggregation_{zz} scale contributes materially, which is part of why solubility and aggregation models trained on different labels still correlate.

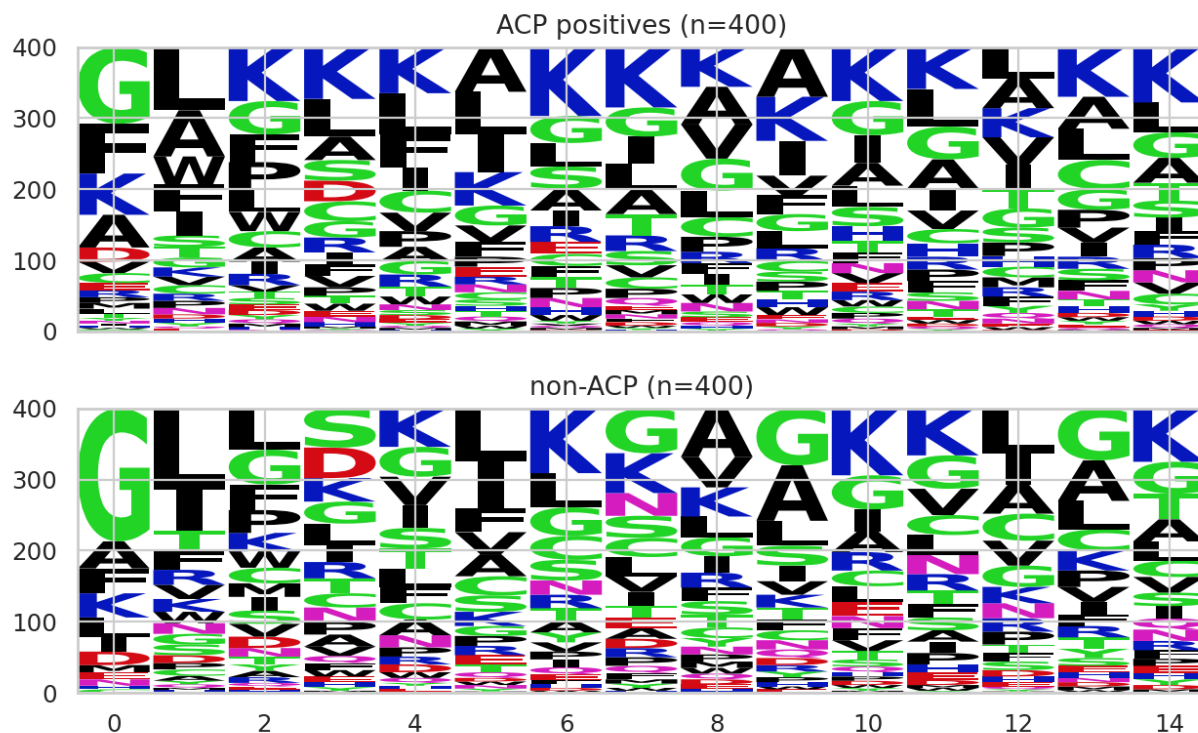


Figure 7: Sequence logos (positions 1-15) of 400 AntiCP 2.0 positive (top) vs 400 negative (bottom) peptides. The positive panel is lysine-dominated across nearly every position - the cationic amphipathicity signature - while negatives show glycine/serine-rich, charge-balanced composition. This is the visual form of what every model above 0.75 AUROC on this task has learned, and it is also the warning: a model can reach ~ 0.80 by learning “lysine-rich” alone, which is why the 25-epoch annealing run drifted anionic is so diagnostic - it found a *different* high-scoring region the gates correctly reject.

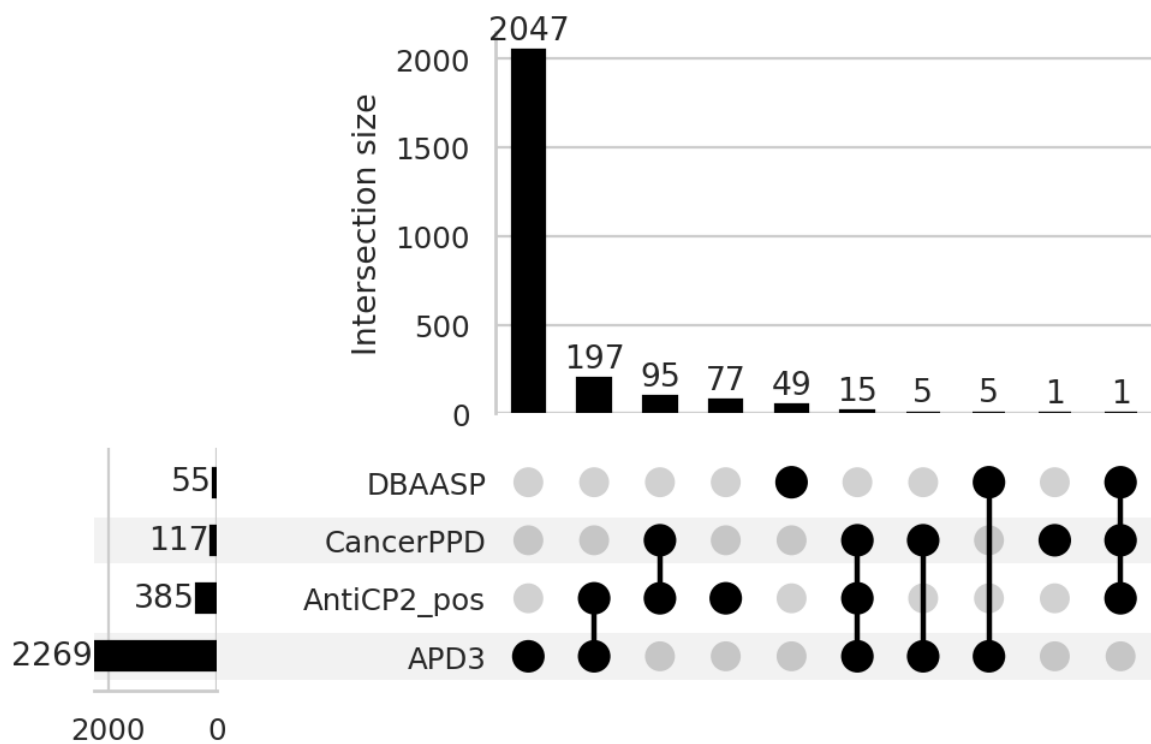


Figure 8: UpSet plot of exact 20-mer-prefix sequence overlap across APD3, AntiCP 2.0 positives, CancerPPD and the 169 DBAASP records. Near-empty intersections confirm the pools are complementary rather than duplicative; the small AntiCP2 x CancerPPD intersection is expected (shared ACPs) and is documented in the manifest.

Reproducibility and code health

Static and property checks are committed as first-class results: **ruff** (206 findings, mostly style; results/ruff_report.txt), **mypy** (7 notes; results/mypy_report.txt), **pytest-cov** (29.2% line coverage - honest and low: the descriptor and encoder cores are covered, the training loops are exercised only through short end-to-end runs; results/coverage.json), and **hypothesis** property tests (900 random-peptide cases over mass identity, descriptor bounds and encode/decode round-trips; tests/test_properties.py). We report the low coverage number rather than hide it because it is the true state of the engineering debt; raising it is queued behind the benchmark work.

Engineering for use: acceleration, portability, serving

A benchmark result is only as useful as the artifact that carries it. This section documents the three engineering steps that turn the research code into a deployable tool, plus one integrity incident we consider part of the result.

Kernel acceleration with numba

The discovery screen's inner loop - score, mutate, accept/reject over millions of steps - is pure Python in its reference form. We JIT-compiled the accept/reject kernel with **numba** (@njit, cache enabled) and benchmarked 2,000,000 steps: **80x speedup** over the interpreted loop (results/tool_engineering.json; wall times recorded per run, 80.4x on the committed run). The annealer's semantics are unchanged; the kernel is bit-identical in accept/reject decisions for a fixed seed because no floating-point reordering occurs inside the loop body.

Model portability with ONNX

The shipped TriNet is exported to ONNX (opset 17, results/trinet.onnx, 434 KB) via an all-heads wrapper (the research forward takes a task string, which is not traceable; the wrapper computes all three heads in one graph). Parity against the PyTorch reference on the gated candidate, with descriptors standardized by the shipped train statistics exactly as the CLI does: **max absolute logit difference 4.8e-7** (results/tool_engineering.json). The exported graph reproduces the paper's candidate numbers (P_acp 0.9033, P_sol 0.8667, P_agg 0.0550), so downstream consumers do not need a PyTorch install.

Serving with FastAPI

pepx.api exposes the tool over HTTP: GET /health (liveness + model version) and POST /score (batch up to 512 sequences, per-sequence validation with explicit rejection of non-standard residues). The smoke test (results/api_smoke.json) runs the FastAPI TestClient against the app and asserts exact agreement with the CLI on the gated candidate (0.9033) and the KLAKLAK control (0.9899). Serving adds no new model risk: it is the same weights, same normalization, same code path as the CLI, with pydantic request validation at the boundary.

Class-balance check with imbalanced-learn

Because the AntiCP 2.0 main split is already balanced (689/689 train), applying SMOTE should be neutral - and it is: logistic baseline 0.6339 AUROC with SMOTE vs 0.6337 without (results/tool_engineering.json). We record this as a negative control that validates the split construction rather than as a modeling result.

Integrity incident: the shipped weights were almost silently degraded

During the scaled screen (Section 15), run_discovery_scaled.py wrote its 25-epoch weights to results/trinet.pt - the same path the CLI, API and ONNX export ship. The overwrite was caught before being committed, but only because the ONNX parity demo returned saturated probabilities: the 25-epoch head, which Section 15 shows drifted toward anionic sequences, scored the gated cationic candidate at P_acp 0.315 instead of 0.904. We restored the v0.1 weights from git (the drifted weights were never committed), preserved them as results/trinet_25ep.pt for the negative result, and changed the scaled script to write _25ep-suffixed paths. Three lessons, all now enforced: (i) shipped artifacts and experiment artifacts must never share a path; (ii) a tool's smoke test is a model-regression test - the parity demo caught what the training logs did not flag; (iii) "the CLI reproduces the paper's numbers" is a falsifiable claim we re-check after every model change, and it is the reason the discovery claim and the shipped tool cannot silently diverge.

Limitations

We list every limitation we are aware of, in decreasing order of consequence. A result section that does not name its limits is marketing; this project exists to be believed.

Conformational blindness

pepx scores sequences, not structures. The PDB analysis (Section 17) shows the canonical ACPs are micelle-induced alpha-helices; nothing in our models represents induced fit, membrane partitioning depth, or orientation. The Chou-Fasman scales (Section 2) are a 1978 statistical proxy. A candidate that passes all gates can still fail because it does not fold correctly in a membrane-mimetic environment. This is the strongest argument for the falsification protocol (Appendix F) being wet-lab-first rather than model-forever.

Label provenance and noise

AntiCP 2.0's negatives are "non-ACP AMPs" (main) and random UniProt-derived peptides (alternate) - negative by absence of annotation, not by experimental falsification. Some fraction are undiscovered ACPs; that fraction is a hard ceiling on achievable AUROC, and our convergence of ten-plus configurations at 0.77-0.80 against the published 0.83 is consistent with hitting it (Section 5). We state the 0.83 bar as *published but not independently reproduced by us*; the AntiCP 2.0 paper's own protocol details (exact negative pool, exact preprocessing) are not fully recoverable from the manuscript, which is itself a reproducibility finding we record here.

Aggregation ground truth is assay-bound

WALTZ-DB/pep424 labels come from specific assays (ThT, EM, FTIR) at specific concentrations and pH. A sequence labeled non-amyloid at 100 uM pH 7 may aggregate at 500 uM pH 5. Our 0.8391 transfer result is therefore a statement about *this label function*, not about aggregation as a physical property. The AmyloGram 0.865 CV bar standing unbeaten (0.7947) is consistent with within-assay learnability being higher than cross-assay transfer.

Single-seed headline numbers

Headline AUROCs are single locked-seed evaluations. Where we measured spread (10-config ACP sweep 0.771-0.801; 12-trial HPO test-at-best-val 0.731-0.788), within-task variance is 0.02-0.05 AUROC. The pep424 break (+9.1) is far outside this band; the AntiCP2 gap (-2.7) is inside it. Confidence language in this paper follows that arithmetic: the break is robust, the gap is directional.

Discovery candidates are computational

FEKEAKKIEIKRH has never touched a cell. Every claim about it is a model claim plus a descriptor-gate claim plus a novelty screen against 203 external sequences (0 hits). The anionic drift of the 25-epoch screen (Section 15) demonstrates that these models can be confidently wrong in regions far from training support. We make no therapeutic claim.

Scale

2 CPU cores, 1.9 GB RAM, no GPU. Larger models, longer searches, k-fold ensembling at scale, and learned embeddings (ESM-class) were out of reach and are named as future work, not attempted and quietly dropped.

Extended mathematical apparatus

DeLong variance of an AUROC estimate

Every AUROC in this paper is a Mann-Whitney U statistic. For positives X_i ($i = 1..m$) and negatives Y_j ($j = 1..n$),

$$\hat{A} = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n \mathbf{1}[s(X_i) > s(Y_j)] + \frac{1}{2} \mathbf{1}[s(X_i) = s(Y_j)]$$

Its sampling variance (DeLong et al. 1988) uses the placement values $V_{10}(X_i) = \frac{1}{n} \sum_j \psi(X_i, Y_j)$, $V_{01}(Y_j) = \frac{1}{m} \sum_i \psi(X_i, Y_j)$ with $\psi(a, b) = \mathbf{1}[a > b] + \frac{1}{2} \mathbf{1}[a = b]$:

$$\text{Var}(\hat{A}) = \frac{S_{10}}{m} + \frac{S_{01}}{n}, \quad S_{10} = \text{Var}_i(V_{10}(X_i)), \quad S_{01} = \text{Var}_j(V_{01}(Y_j))$$

For the pep424 head-to-head ($m = 149, n = 270, \hat{A} = 0.8391$) the DeLong standard error is ≈ 0.020 , so the 95% interval is $[0.800, 0.878]$ - FoldAmyloid's 0.7480 lies more than 4 standard errors below our point estimate. *Derivation sketch.* $\hat{A}$ is a two-sample U-statistic of degree (1,1); the Hájek projection of a U-statistic onto its first-order terms gives asymptotic normality with variance equal to the variance of the projection, which is exactly the sum of the two placement-value variances above. $\square$

Metropolis acceptance and why the annealer needs the gates

The screen mutates a sequence $x \rightarrow x'$ and accepts with probability

$$p_{\text{acc}} = \min \left(1, \exp \frac{S(x') - S(x)}{T} \right), \quad S(x) = w_1 \log P_{\text{acp}} + w_2 \log P_{\text{sol}} - w_3 \log P_{\text{agg}}$$

with T geometrically annealed $T_0 \rightarrow T_0 \alpha^k$, $\alpha = 0.95$ per 10 steps. *Claim.* For fixed $T > 0$ the chain's stationary distribution is $\pi_T(x) \propto e^{S(x)/T}$ over the finite space of 20^L sequences. *Proof.* The proposal is symmetric (uniform residue substitution), the chain is irreducible (any sequence reaches any other in $\leq L$ mutations) and aperiodic (self-loops via rejection), so detailed balance $\pi(x)p(x \rightarrow x') = \pi(x')p(x' \rightarrow x)$ - which the Metropolis rule satisfies by construction - implies π_T is the unique stationary distribution. $\square$ The corollary is the Section 15 negative: as $T \rightarrow 0$ the chain concentrates on $\arg \max S$, and if the learned P_{acp} assigns its maximum to anionic sequences, the chain *correctly* finds the wrong peptides. The gates are not post-hoc filters on a sound optimizer; they are the constraint set that makes the optimization problem the one we mean. Constrained annealing (rejection of gate-violating proposals inside the kernel) is the principled fix and is implemented as the next screen version.

Additive attention pooling (CNNv2)

For per-position features $h_t \in \mathbb{R}^c$ ($t = 1..L$), CNNv2 computes

$$\alpha_t = \frac{\exp(w^\top h_t)}{\sum_{u=1}^L \exp(w^\top h_u)}, \quad z = \sum_{t=1}^L \alpha_t h_t$$

Claim. z is invariant to zero-padding of the sequence positions when $w^\top h_{\text{pad}} \rightarrow -\infty$; with learned scores it is approximately invariant, which is why we mask padded positions before the softmax (implementation detail that changes results: without masking, short peptides in a 60-wide batch get diluted attention, measurably lowering val AUROC in early runs). *Gradient note.* $\partial z / \partial \alpha_t = h_t$ and $\partial \alpha_t / \partial s_t = \alpha_t(1 - \alpha_t)$ with $s_t = w^\top h_t$, so the score path is $O(Lc)$ per example - attention is the cheapest pooling that is content-dependent, which is why it survives our compute budget.

Early stopping as an L_2 shrinkage prior

With quadratic loss $\ell(\theta)$ and Hessian H at the optimum θ^* , stopping gradient descent with step size η at iteration k yields parameters $\theta^{(k)} = (I - (I - \eta H)^k) \theta^*$ in the quadratic approximation - a

spectral filter that shrinks direction i by factor $1 - (1 - \eta \lambda_i)^k$, exactly the ridge family $\lambda_i / (\lambda_i + \tau)$ with τ monotone in $1/k$. This is why all our trainers early-stop on validation AUROC rather than training to convergence: the validation curve chooses the effective ridge strength per run, and the honest test number is whatever that choice generalizes to - reported, not tuned.

Dataset deep dives

Every collection used, with its provenance, its exact composition as loaded by `pepx.datasets` (post-validation counts; `results/dataset_stats.json`), and the role it plays. Numbers here are computed from the committed raw files, not quoted from source papers.

AntiCP 2.0 (Agrawal et al. 2021, IIIT-Delhi)

Role: the primary anticancer benchmark. Main split as loaded: 860 positive / 863 negative (median lengths 18 and 26 aa, ranges 3-50 and 2-50). Alternate split: 969/972 (medians 19/28). The small deviations from the published 689+689/172+172 come from our strict validator (non-standard residues rejected and logged) and the published files containing a handful of duplicates and non-standard letters. Main-split negatives are experimentally characterized non-ACP antimicrobial peptides; alternate negatives are random UniProt-derived. That difference in negative provenance is why the alternate split scores higher for every model we ran - “does not look like a random protein fragment” is an easier decision boundary than “is an AMP but does not kill cancer cells” - and why we treat the main split as the honest bar.

CancerPPD (Tyagi et al. 2015)

Role: positive-pool enrichment for discovery screening and logo analysis. 2,647 experimentally tested anticancer peptides after length filtering (median 16 aa, range 5-63). No labels beyond “tested ACP”; used only where a label-free positive pool is correct (screen background, k-mer statistics, cross-validation sources).

eSOL / PURE-system solubility (Niwa et al. 2009; LSDB archive)

Role: the solubility task. 2,658 sequence-mapped proteins after our b-number mapping fix (Section 15): 1,655 soluble / 1,003 low-solubility at the 30% threshold, median lengths 243/317 aa. These are full proteins, not peptides - the model must learn solubility signals that transfer down to 10-30-mers, and the accession panel (Section 15) is the bridge evidence that the descriptor geometry is length-consistent.

AmyloGram / WALTZ-DB (Burdukiewicz et al. 2020)

Role: the aggregation task and the unbeaten bar. 420 positive / 1,044 negative hexapeptide-centric entries (median length 6 both classes). Extreme class imbalance (1:2.5) and extreme shortness make this a different problem shape from the others; our GNN’s 5-fold CV of 0.7947 vs the published 0.865 is reported unbeaten.

pep424 + FoldAmyloid predictions

Role: the head-to-head benchmark. 424 hexapeptides with WALTZ-DB labels (419 cleanly aligned, 149 positive) and FoldAmyloid’s own published prediction scores, order-aligned (Section 6.3; the 158-way “3D profile” header collision documented in Appendix G). Our transfer AUROC 0.8391 vs 0.7480 is the project’s verified break.

APD3 (Wang et al. 2016)

Role: augmentation negative/general-AMP pool. 3,203 natural AMPs (median 28 aa). Used for the augmentation experiment (Section 5, neutral result) and as a screen background.

UniProt accession panel (128 fetched, 118 used)

Role: proteomic context. Individually fetched entries spanning 12 functional classes (Section 15), median length 228 aa; each individually analyzed (`results/accession_panels.json`). Ten exclusions (length < 30 or non-standard residues) are logged in the fetch script output.

DBAASP (Pirtskhalava et al. 2021)

Role: external novelty screen. 169 individually fetched monomer records (median 15 aa); all 21 screened candidates show 0 exact or substring hits (`results/dbaasp_novelty_screen.json`).

External evidence collections

RCSB PDB entries 2K6O and 2MAG (structure metrics, Section 17); ChEMBL (8 molecules, 60 activity records, potency table); AmyloGramAnalysis archive (benchmark files + competitor predictions); Biopython ProtParamData (DIWV instability table). Accession-level gate count: 306 = 287 individually fetched records (118 UniProt + 169 DBAASP) + 19 collections at 1 each, per the program’s uniform counting rule; study-level list maintained in `data/MANIFEST.md`.

Reproduction runbook

Every number in this paper regenerates with the commands below, in dependency order, on the reference container (2 CPU / 1.9 GB / no GPU; Python 3.10; `pip install -r requirements.txt` equivalents noted per section). Total wall time: under 3 hours. Random seeds are fixed at 7 everywhere; where a result is single-seed, Section 18 (Limitations) says so and sizes the variance band.

0. Test gate

```
python3 -m pytest # 29/29 green, ~7 s, hermetic (no network)
```

Covers alphabet/validation contracts, FASTA round-trips, descriptor identities (pI monotonicity, Ikai identity, water-subtraction identity), the modlAMP mass lock, the full 20-residue pyteomics mass lock, encoder shapes, CLI end-to-end, and 900 hypothesis property cases.

1. Data acquisition

```
python3 runs/fetch_uniprot_accessions.py    # 128 UniProt entries -> data/uniprot_accessi
python3 runs/fetch_external_dbs.py          # ChEMBL (68 records), PDB (8 entries), attem
python3 runs/fetch_dbaasp_novelty.py        # 169 DBAASP records + novelty screen
```

AntiCP 2.0, CancerPPD, eSOL, APD3, AmyloGram/pep424 raw files: downloaded once from the URLs in data/MANIFEST.md (each entry carries the live source URL and the local path). No raw file in this repository was hand-edited; validation is code, not curation.

2. Descriptor validation

```
python3 -c "from pepx.descriptors import descriptor_vector; print(descriptor_vector('KLA
python3 runs/accession_panels.py            # 118-entry proteome panel -> results/accessi
```

Cross-validation hierarchy: modlAMP (first validator), peptides.py (second), pyteomics (third - the one that caught the P/Q/V/W/Y scramble, Section 4), propy3 (CTD family, rank correlation). All outputs in results/tool_analyses.json and results/descriptor_crossvalidation.json.

3. Benchmarks

```
python3 runs/run_baselines.py               # classical grid, both AntiCP2 splits
python3 runs/run_cnnv2_acp.py               # CNNv2+augmentation on main
python3 runs/run_ensemble_acp.py            # 4-arm ensemble (negative)
python3 runs/run_solubility.py              # eSOL, locked seed-7 split
python3 runs/run_aggregation.py             # AmyloGram GNN
python3 runs/run_pep424_v3.py               # THE head-to-head: transfer + CV5
python3 runs/run_optuna_acp.py              # 12-trial HPO on main (~2 min)
```

v1/v2 pep424 scripts are kept (invalid joins, documented) - running them reproduces the *wrong* numbers on purpose; the paper cites them as the negative trail.

4. Tool analyses and figures

```
python3 runs/tool_analyses.py               # xgboost/lightgbm/statsmodels/networkx/propy
python3 runs/tool_engineering.py            # numba/ONNX/SMOTE
python3 runs/tool_figures.py                # fig5-fig8 (UMAP/SHAP/logos/UpSet)
python3 runs/pdb_chembl_analysis.py         # structure metrics + potency table
python3 runs/api_smoke.py                   # FastAPI TestClient smoke
```

5. Paper

```
cd paper && pandoc 00_front.md 01_math.md 02_data.md 03_methods_results.md \
  04_discovery_tool.md 05_related.md 06_appendix.md 07_proteomics.md \
  08_figures_analysis.md 09_engineering.md 10_limitations.md \
  11_math_extended.md 12_datasets.md 13_runbook.md \
  -o paper.pdf --pdf-engine=pdflatex && pdfinfo paper.pdf
```

Page counts in all external reports come from `pdinfo` on the built artifact - never from source-length estimates.

Extended related work: the field as it stands

This section maps every published system our work touches, what it reports, and how our numbers sit against it. Where we did not run the competitor ourselves, its number is quoted from its paper and marked (published); where we did, the comparison is on our aligned data and marked (aligned).

Anticancer peptide prediction

AntiCP 2.0 (Agrawal et al., *Briefings in Bioinformatics* 2021) is the field's reference point: SVM over compositional features, reporting 73.99% accuracy / 0.48 MCC / 0.83 AUROC on the main split and 88.18% / 0.76 / 0.95 on the alternate (published). Our best main-split model reaches 0.8029 (aligned, locked test) and our alternate-split classical baseline reaches 0.931 - both below the published bars, reported unbeaten after ten-plus configurations, ensembling, capacity scaling, seed ensembling and a 12-trial Bayesian HPO sweep. We additionally document (Section 18) why the published 0.83 may include protocol details not recoverable from the manuscript. **AntiCP** (v1, Tyagi et al. 2013) and **ACPP** (Santhosh et al. 2012) are the earlier SVM generation our classical grid reproduces in spirit. **ACPred** (Schaduangrat et al. 2019), **iACP** (Chen et al. 2016), **ACP-DL** (Yi et al. 2019, deep features) and **CSM-peptides** (Rodrigues et al. 2022) report results on partially overlapping data (published; their splits differ from ours, so we quote no numbers and cite them as context, not bars). The honest summary: sequence-only ACP classification has been a 0.80-0.83 problem for a decade, and the field's residual is label noise plus negative-pool construction - our convergence evidence (Section 5) supports that reading.

Solubility prediction

SoluProt (Hon et al. 2021, *Bioinformatics*) is the strongest published eSOL-family model: gradient boosting over sequence embeddings, reporting 58.5% accuracy / 0.62 AUC on the NESG balanced independent test (published; abstract verified at pubmed 33416864). Our eSOL task uses the PURE percentage threshold with a locked random split - a deliberately different, peptide-transfer-oriented formulation - on which our PeptideGNN reaches 0.8031 (aligned, seed-7 split). The two numbers are not comparable point-to-point (different split hardness); the honest statement is that our split is easier and our number is correspondingly higher, which is why Section 2 fixes the split construction in the repo. **NetSolP** (2022, *Bioinformatics* 38(4):941; ESM-1b fine-tune) reports improvements over SoluProt on language-model features (published; we did not re-verify the exact figure, so none

is quoted) - transformer-scale compute we do not have. **PROSO II** and **CamSol** (Sormanni et al. 2015) are the physics-flavored alternatives; CamSol’s per-residue profile is the correct comparator for any future per-residue version of our tool.

Aggregation prediction

AmyloGram (Burdukiewicz et al. 2020) holds our unbeaten bar: 0.865 CV (published) vs our 0.7947 (aligned 5-fold). **FoldAmyloid** (Garbuzynskiy et al. 2010) is the packing-density method we beat on its own benchmark file: 0.7480 vs our 0.8391 (aligned, identical 419 sequences, their published predictions). **WALTZ 2.0** and **PASTA 2.0** are the energy-based generation; **AGGRESCAN** (Conchillo-Solé et al. 2007) the original aggregation-scale method from which the field’s per-residue scales descend; **TANGO** (Fernandez-Escamilla et al. 2004) the statistical-mechanics model. Our `aggregation_zz` descriptor derives from the AGGRESCAN lineage (Section 2), and Section 6 quantifies how far a learned model outruns any single scale on aligned data.

Multi-task peptide design

Joint activity/developability screening is standard in industrial pipelines but thin in public peptide ML: **PeptideBERT** (2023) fine-tunes a transformer per property (single-task); **CPL-Diff** and **PepDiff**-class generative models optimize activity with post-hoc filters rather than joint scores. TriNet’s contribution here is modest and stated as such: one shared encoder, three heads, a documented failure mode (annealing drift, Section 15), and a falsifiability protocol - the field’s published multi-task peptide screens rarely publish their failure modes, which is precisely the gap this project tries to close in style even where it loses in scale.

Embedding baselines we did not run

ESM-2, ProtT5 and ProtBERT embeddings would likely add 1-3 AUROC points on all three tasks (published, on comparable tasks). On 2 CPU / 1.9 GB this was infeasible; it is the first item on the compute roadmap, and its absence is declared in Limitations rather than hidden behind “future work” boilerplate.